



IT-988GSE

High Tg / Halogen Free / Lead Free / Ultra Low Loss Laminate & Prepreg

@ 10GHZ

- 56Gbps per channel / NRZ & PAM4
- Advanced Resin Technology
- Low Dk (<3.24) & Ultra Low Df (<0.0014)
- Excellent Insulation & CAF Resistance
- Very Stable Dk – Df across Frequency

Laminate properties

Items	IPC TM-650	Typical Value	Unit
Peel Strength, minimum			
A. 0.5 oz Very low copper roughness	2.4.8	3.50	PLI
Volume Resistivity	2.5.17.1	4.0x10 ¹⁶	MΩ-cm
Surface Resistivity	2.5.17.1	1.37x10 ¹²	MΩ
Moisture Absorption, maximum	2.6.2.1	0.26	%
Permittivity (Dk, 53% resin content)			
A. 1 GHz / 2 GHz	2.5.5.13	3.24 / 3.24	--
B. 5 GHz / 10 GHz		3.24 / 3.24	
C. 15 GHz / 20 GHz		3.24 / 3.24	
Loss Tangent (Df, 53% resin content)			
A. 1 GHz / 2 GHz	2.5.5.13	0.0013 / 0.0014	--
B. 5 GHz / 10 GHz		0.0014 / 0.0014	
C. 15 GHz / 20 GHz		0.0014 / 0.0014	
Flexural Strength, minimum			
A. Length direction	2.4.4	TBD	N/mm ²
B. B. Cross direction		TBD	
Thermal Stress 10 s at 288°C			
Unetched / Etched	2.4.13.1	Pass / Pass	Rating
Flammability	UL94	TBD	Rating
Glass Transition Temperature (Tg)			
A. TMA / DSC / DMA	2.4.25	190 / 190 / 205	°C
Decomposition Temperature	2.4.24.6	405	°C
X/Y Axis CTE (40°C to 125°C)	2.4.24	15/16	ppm/°C
Z-Axis CTE			
A. Alpha 1 & 2	2.4.24	58 / 325	ppm/°C
B. 50 to 260 Degrees C		2.95	%
Thermal Resistance			
A. T288	2.4.24.1	> 120	Minutes
B. T300		> 120	Minutes
UL CTI		Under testing	
UL MOT (Maximum operating temperature)		Under testing	
Thermal conductivity		Under testing	
IPC Slash sheet	4101	TBD	

